

ELMI CV

Əmirullah Məmmədov, DSc, PhD

Emeritus professor və aparıcı elmi işçi

Nanotexnologiya Tədqiqat Mərkəzi (NANOTAM), Bilkent Universiteti, Ankara, Türkiyə

Əlaqə məlumatları və identifikatorlar

Tam ad (nəşrlərdə yazıldığı kimi)	Amirullah M. Mamedov
Azərbaycan dilində ad	Əmirulla Məmmədov
Yaşadığı ölkə	Türkiyə
Rəsmi e-poçt	mamedov@bilkent.edu.tr
Elm sahəsi	Optika; fotonika; nanotexnologiya

Profil xülasəsi

Əmirullah Məmmədov 2011-ci ildən Bilkent Universitetinin Nanotexnologiya Tədqiqat Mərkəzində emeritus professor və aparıcı elmi işçidir. Nəşrlərdə adı Amirullah M. Mamedov yazılır. 1973-cü ildə İoffe İnstitutunda fizika və riyaziyyat üzrə PhD dərəcəsi alıb. Dissertasiya fotoseqnetoelektrik materialların optik və elektrooptik xassələrinə aiddir və Vavilov Dövlət Optika İnstitutunda yerinə yetirilib. 1982-ci ildə Riqada fizika və riyaziyyat üzrə elmlər doktoru, 1988-ci ildə optika üzrə professor elmi adını alıb. CV 350-dən çox nəşr göstərir. 1992–2006-cı illərdə ASELSAN-ın elmi məsləhətçisi olub.

Təhsil

İllər	Dərəcə	Sahə / dissertasiya mövzusu	Təhsil müəssisəsi, şəhər, ölkə
1988	Optika üzrə professor		SSRİ Ali Attestasiya Komitəsi, Moskva
1982	DSc	Fizika və riyaziyyat. Habilitasiya. Riqada verilib	SSRİ Ali Attestasiya Komitəsinin regional şurası, Rıqa, Latviya
1977	Fizika üzrə dosent	Bərk cisim fizikası	USSR
1973	PhD	Fizika və riyaziyyat. Fotoseqnetoelektrik materialların optik və elektrooptik xassələri. İş Vavilov Dövlət Optika İnstitutunda görüleb	İoffe Fizika-Texnika İnstitutu, Sankt-Peterburq, Rusiya

Elmi və peşəkar vəzifələr

İllər	Vəzifə	Kafedra / bölmə	Təşkilat, şəhər, ölkə
2011–hazırda	Emeritus professor və aparıcı elmi işçi		NANOTAM, Bilkent Universiteti, Ankara, Türkiyə
1992–2006	Elmi məsləhətçi	Mikroelektronika, tuşlama və elektrooptika bölməsi	ASELSAN, Turkey
2004–2010	Məsləhətçi		SAVTAG, TÜBİTAK və SATEM, Türkiyə Milli Müdafiə Nazirliyi
	Fizika və elektrik-elektronika mühəndisliyi professoru		Çukurova Universiteti, Adana, Türkiyə

İllər	Vəzifə	Kafedra / bölmə	Təşkilat, şəhər, ölkə
	Professor və qeyri-xətti optika bölməsinin müdiri		Bakı Dövlət Universiteti, Bakı, Azərbaycan

Elmi göstəricilər

Göstərici	Dəyər	Mənbə	Yoxlama tarixi
Beynəlxalq jurnallarda nəşrlər	350-dən çox	CV	

Nəşrlər

Rəylənmiş (peer-reviewed) jurnal məqalələri

1. Modeling and Simulation of the Ferroelectric Based Micro-Gyroscope: FEM Analysis, Ferroelectrics, 447(1), 46, (2013)
2. 2D Anisotropic Photonic Crystals of Hollow Semiconductor Nanorod with Liquid Crystals, Applied Mechanics and Materials, 394(1), 38 (2013)
3. Electron Energy Loss Spectroscopy and the Electronic Band Structure of KNbO₃ Ferroelectric, Ferroelectrics, 461(1), 1 (2014)
4. Mechanical, Electronic and Optical Properties of Bi₂S₃ and Bi₂Se₃ Compounds: first principle investigations, J. Molecular Modeling, 20(4), 2180 (2014)
5. Optical Properties of Metamaterial Based Devices Modulated by a Liquid Crystal, Appl. Phys. A, 117(2) (2014), 611
6. Ferroelectric Based Photonic Crystal Cavity by Liquid Crystal Infiltration, Integrated Ferroelectrics, 158(1), (2014) 1-12
7. Organic semiconductor-based photonic crystals for solar cell arrays: band gap and optical properties, J. Modern Opt. 161 (21), (2014) 1754
8. Band Gap and Optical Transmission in Fibonacci Type One Dimensional A₅B₆C₇ Based Photonic Crystals, Phys. Stat.Sol.(c) 12(6), 540-544 (2015)
9. Dynamic Nonlinear Optical Processes in Some Oxygen-Octahedra Ferroelectrics: first principle calculations, Ferroelectrics, 483(1), 26-42 (2015)
10. Optical Properties of the Narrow-Band Ferroelectrics: first principle calculations, Ferroelectrics 483(1), 43-52 (2015)
11. Influence of Crown Margin Design on the Stress Distribution in Maxillary Canine Restored by All-ceramic Crown: A Finite Element Analysis, J. Korean Dent Sci., 8(1),28-35 (2015)
12. BaTiO₃ and TeO₂ based gyroscopes for guidance systems: FEM analysis, Ferroelectrics, 497(1), 15-23 (2016)
13. Phononic band gap and wave propagation on polyvinylidene fluoride-based acoustic metamaterials, Cogent Physics, 2, 1169570 (2016)
14. Topological Insulator Based Locally Resonant Phononic Crystals: Wave Propagation and Acoustic Band Gaps, Ferroelectrics, 499(1) 123-129 (2016)
15. AIBIIICVI₂ (A=Cu, Ag; B=Ga, In; C=S, Se, Te) Based Photonic Crystal Superlattices: Optical Properties" Physica Status Solidi (C), 1600156 (2017)
16. Optical and Magnetic Properties of Some XMnSb and Co₂YZ Compounds: ab initio calculations, Physica Status Solidi (C), 1600182 (2017)
17. Wave Propagation and Acoustic Band Gaps of Two-Dimensional Liquid Crystal/Solid Phononic Crystals, Appl. Phys. A, 123(1):23 (2017)
18. Spectral Scalability and Optical Spectra of Fractal Multilayer Structures: FDTD analysis, Appl. Phys. A, 123(1):33 (2017)
19. Band gap structure of elliptic rods in water for a 2D phononic crystal, Appl. Phys. A, 123 (3): 212 (2017)
20. Topological Insulators: Electronic Band Structure and Spectroscopy, J. Physics, Materials Science and Engineering, 175 (2017) 012004

21. Fracture of Femoral Neck: Analysis of New Implant Models with and without Slit by the FEM, Azerbaijan Medical Journal, iss.3, 12 (2017)
22. A5B6C7 Ferroelectrics as Novel Materials for Phononic Crystals, Ferroelectrics, 511 (1) 12, 2017
23. Bud Sümüyü Boynu Sınığı: Dəlikli və Dəliksiz Yeni Implant Modellerinin Sonlu Elementlər Üsulu ilə Təhlili, Azərbaycan Tibb Jurnalı 2017(3) s. 12
24. Fibonacci Sequences Quasiperiodic A5B6C7 Ferroelectric Based Photonic Crystal: FDTD analysis, Integrated Ferroelectrics, 2017
25. Wave Propagations in Metamaterial Based 2D Phononic Crystal: Finite Element Analysis, J. Physics, Materials Science and Engineering, 1045 (2018) 012034
26. Stress Distribution of a New Type of Perforated Implants in Femur Neck Fracture: Finite Element Model, J. Physics: Materials Science and Engineering, 1045 (2018) 012017
27. Electronic band structure of rare-earth ferroelastics: theoretical investigations, JOAM 20 (12) 69, 2018
28. Electronic, Mechanical, and Optical Properties of Ruddlesden-Popper Perovskite Sulfides: first principle calculation, Ferroelectrics, 535(1), 142 (2018)
29. Multiple Band gaps of a Ferroelectric Based 2D-Phononic crystal slab, Ferroelectrics, 535(1) 152 (2018)
30. Phononic Band Gap and Wave Propagation on multiferroic based acoustic metamaterials, Ferroelectrics, 2019, 539 (1) 84
31. Photonic Band Gap of Multiferroic-Dielectric Materials in the IR region: FDTD method, Ferroelectrics, 2019, 539(1) 50
32. Electronic properties of spin excitation in multiferroics with a spinel structure: First principles calculation, Ferroelectrics, 2019, 539 (1) 41
33. Electronic and Elastic Properties of the Multiferroic Crystals with the Kagome Type Lattices -Mn₃V₂O₈ and Ni₃V₂O₈: First Principle Calculations, Ferroelectrics, 544(1), 11, 2019
34. Two-Dimensional Phononic Band structure of Archimedean-Logarithmic Spiral- Based Slabs Ferroelectrics, 544(1), 112, 2019
35. BaTiO₃ Based Photonic Time Crystals and Momentum Stop -Bands, Ferroelectrics, 2020 557(1), 105
36. Complete Photonic Band Gaps in Sn₂P₂X₆ (X=S, Se) Supercell Photonic Crystals, Ferroelectrics, 2020 557(1), 92
37. Band structure and transmission spectra in multiferroic based Sierpinski- carpet phononic crystal, European Physical J.- Applied Physics, 2020, 90(2)
38. Phononic Crystals with Archimedean-like Tiling: Band Structure and the Transformation of Sound, Integrated Ferroelectrics, 2021, 220(1)
39. Slater insulator phase of X₂ (X = Na, Li)IrO₃: First principles calculation, Integrated Ferroelectrics 2021, 220(1)
40. Second Harmonic Generation in Generalized Ferroelectric Superlattices, Integrated Ferroelectrics, 2021, 220(1)
41. Band structures of metacomposite based phononic crystals in quasi- Sierpinski fractals, EPITOANYAG-JOURNAL OF SILICATE BASED AND COMPOSITE MATERIALS, 2022, 74 (2)
42. Observation of shear bond strength between zirconia core and silica based composite material; a finite element analysis, EPITOANYAG-JOURNAL OF SILICATE BASED AND COMPOSITE MATERIALS, 2022, 74 (2)
43. SECOND HARMONIC GENERATION IN LOCALLY RESONANT NONLINEAR WEYL SEMIMETAL BASED ACOUSTIC METAMATERIALS: FINITE ELEMENT ANALYSIS, AJP, Fizika (C) 2023, 77
44. DIRAC CONE AND TOPOLOGICAL STATES IN 2D PHONONIC CRYSTALS, AJP, Fizika (C) 2023, p.71
45. TUNABLE PHONONIC WAVEGUIDE: FINITE ELEMENT ANALYSIS, AJP, Fizika (C) 2023,
46. Topological semimetals in the ZrB₅ (B = Se, Te) material class: Weyl points and electronic properties, Ferroelectrics, 2024, 618 (4)
47. Observation of elastic Weyl points in ferroelectric based sonic metamaterials Ferroelectrics, 2024, 618 (4)
48. The structural, elastic, electronic, and optical properties of Li₂RuO₃ compound in different structures, Chinese Journal of Physics, 2024, 90, 1026

49. The structural, elastic, electronic, and optical properties of the rhombohedral $\text{La}_2\text{NiMnO}_6$ double perovskite compound at different temperatures, *Comp. Cond. Matt.*, 2024, 41, e00984.
50. The structural, mechanical, electronic, and optical properties of multiferroic LiCu_2O_2 under different pressures, *Bulletin of Materials Science*, 2025, 48:6 Conferences (more than 50 international conferences, meetings and workshops; below are some conferences from 2012 up to now): inv. presentations Bilkent University
51. 21-st International Symposium on Applications of Ferroelectrics, ISAF2012, (2012, Portugal, Aveiro)
52. 11-th European Conference on Applications of Polar Dielectrics, ECAPD-11, (2012, Portugal, Aveiro)
53. 4-th Conference Piezoresponse Force Microscopy and Nanoscale Phenomena in Polar Materials, (2012, Portugal Aveiro)
54. International Ultrasonic Symposium (2013, Czech Republic, Prague)
55. International Symposium on the Applications of Ferroelectrics (2013, Czech Republic, Prague)
56. 13-th International Meeting on Ferroelectricity (2013, Poland, Katowice – Krakow)
57. International Conference “Novel Semiconductor Materials and Structures (2013, Azerbaijan, Baku)
58. International Conference on Mechatronics, Applied Mechanics and Energy Engineering - MAMEE 2013 (2013, Singapur).
59. 5th International Conference on Metamaterials, Photonic Crystals and Plasmonics, META – 2014, (2014, Singapur)
60. 19th International Conference on Ternary and Multinary Compounds, ICTMC19, (2014, Niigata Japan)
61. 12th Russia/CIS/Baltic/Japanese Symposium on Ferroelectricity, RCBJSF-2014, (2014, Riga Latvia)
62. 9th International Conference on Functional Materials and Nanotechnologies, 2014 –FM&NT, (2014, Riga, Latvia)
63. Science and Application of Thin Films, Conference and Exhibition, SATF2014, (2014, Izmir, Turkey)
64. International Organic Electronic Material Technologies Conference, OEMT 2015 (Elazığ, Turkey; 25-28 March, 2015)
65. 13-th European Meeting on Ferroelectricity, EMF-2015 (Porto, Portugal; 28 June-03 July, 2015)
66. 13th Russia/CIS/Baltic/Japanese Symposium on Ferroelectricity, RCBJSF2016, (2016, Shimane, Japan)
67. 7th International Conference on Metamaterials, Photonic Crystals and Plasmonics, META – 2016, (2016, Malaga, Spain)
68. 20th International Conference on Ternary and Multinary Compounds, ICTMC- 20, (2016, Halle, Germany)
69. 4th International Conference on Competitive Materials and Technology Processes, IC-CMTP 4, (2016, Miskolc-Lillafüred, Hungary)
70. The 1st International Symposium on Intelligent and Smart Materials. (2016, Miskolc-Lillafüred, Hungary)
71. 6th International Conference on Materials and Applications for Sensors and Transducers – IC-MAST (2016, Athens, Greece)
72. 2nd International Congress on the World of Technology and Advanced Materials - WITAM 2016 (2016, Kirshehir, Turkey)
73. 14-th International Meeting on Ferroelectricity (IMF 2017, San Antonio, USA)
74. 3rd International Conference on Rheology and Modelling of Materials (ICRMM3, 2017, Miskolc-Lillafüred, Hungary).
75. 14th Russia/CIS/Baltic/Japanese Symposium on Ferroelectricity, (RCBJSF-2018, St. Petersburg, Russia)
76. 14-th European Conference on Applications of Polar Dielectrics, (ECAPD2018, Moscow, Russia)
77. 4th International Conference on Competitive Materials and Technology Processes, (IC-CMTP 5 2018, Miskolc-Lillafüred, Hungary)
78. The European Federation of National Associations of Orthopaedics and Traumatology Congress (EFORT 2018, Barselona, Spain)
79. 14-th European Meeting on Ferroelectricity, EMF-2019 (Lausanne, Switzerland; 15 July-19 July, 2019)
80. 10th International Conference on Metamaterials, Photonic Crystals and Plasmonics, META – 2019, (Lisbon, Portugal; 23-26 July 2019)

81. 4th International Conference on Rheology and Modelling of Materials (IC- RMM4, 2019, Miskolc-Lillafüred, Hungary).
82. 36-th International Physics Congress of the Turkish Physical Society (TPS-36, September, 1- 5, 2020. Istanbul, Turkey)
83. International Conference “Functional Materials & Nanotechnologies” (FM&NT 2020, November 23-26, 2020. Vilnius, Lithuania)
84. 6th International Conference on Competitive Materials and Technology Processes (IC-CMTP 6 October 4-8, 2021, Miskolc-Lillafüred, Hungary)
85. 7th International Conference “Modern Trends in Physics” (MTP-7, 15-17 December, 2021, Baku Azerbaijan),
86. 15-th International Meeting on Ferroelectricity (IMF 2023, Tel-Aviv. Israel)
87. 8th International Conference “Modern Trends in Physics” (MTP-8, November, 30 – December 01, 2023, Baku Azerbaijan)
88. Functional Materials Science and Engineering Symposium, (June 11-12, 2024, Mersin, Türkiyə)
89. International Conference on Energy and Environmental Technologies in Engineering and Architecture (ICETEA, 19-21 December 2024, Baku, Azerbaijan)

Mükafatlar, fəxri adlar və təqaüdlər

İl	Mükafat / fəxri ad	Təltif edən qurum
	SSRİ və Azərbaycan dövlət mükafatları	
	Azərbaycanın fəxri elm xadimi	
	Dostluq ordeni	Azərbaycan Respublikası
	Diaspor fəaliyyətində xidmətə görə medal	Azərbaycan Respublikası
2019	Bakı Dövlət Universitetinin 100 illiyi yubiley medalı	Baku State University

Dillər

Dil	Səviyyə
İngilis dili	Səlis
Türk dili	Səlis
Rus dili	Səlis
Azərbaycan dili	Səlis
İvrit dili	Yaxşı